Assignment 3 - Statistical Analysis of NYC Flights 2013

By: Pratheeksha Sumesh - SE24UCAM040

Introduction and Dataset Overview

The **nycflights13** dataset records all flights departing from New York City in 2013 across three airports: Newark (EWR), JFK, and LaGuardia (LGA). It contains **336,776 observations** and **19 variables** covering 16 airline carriers.

Only 4 out of 19 variables are analyzed because the remaining variables are either identifiers (flight number, tail number, time stamps), constants (year is always 2013), or qualitative categories (carrier, origin, destination) which are not suitable for analysis.

Variable Classification:

Table 1: Classification of Key Variables

Variable	**Type**	**Description**
carrier	Qualitative	Airline code (16 levels)
origin	Qualitative	Departure airport (EWR, JFK, LGA)
dest	Qualitative	Destination airport
dep_delay	Quantitative	Departure delay (minutes)
arr_delay	Quantitative	Arrival delay (minutes)
air_time	Quantitative	Time in air (minutes)
distance	Quantitative	Flight distance (miles)

Summary Statistics

Table 2: Summary Statistics for Key Quantitative Variables

Statistic	**dep_delay**	**arr_delay**	**air_time**	**distance**
Minimum	−43	−86	20	17
1st Qu.	−5	−17	82	502
Median	−2	−5	129	872
Mean	12.64	6.90	150.7	1040
3rd Qu.	11	14	192	1389
Maximum	1301	1272	695	4983

Inferences:

- For `dep_delay`, mean (12.64) $\gg$ median (-2), confirming heavy right skew
- Maximum delay of 1,301 minutes ($\approx$21 hours) is a striking outlier
- Most flights arrive early (median = -5 mins) but extreme arrivals push the mean to 6.90

Bar Charts

Flights by Carrier

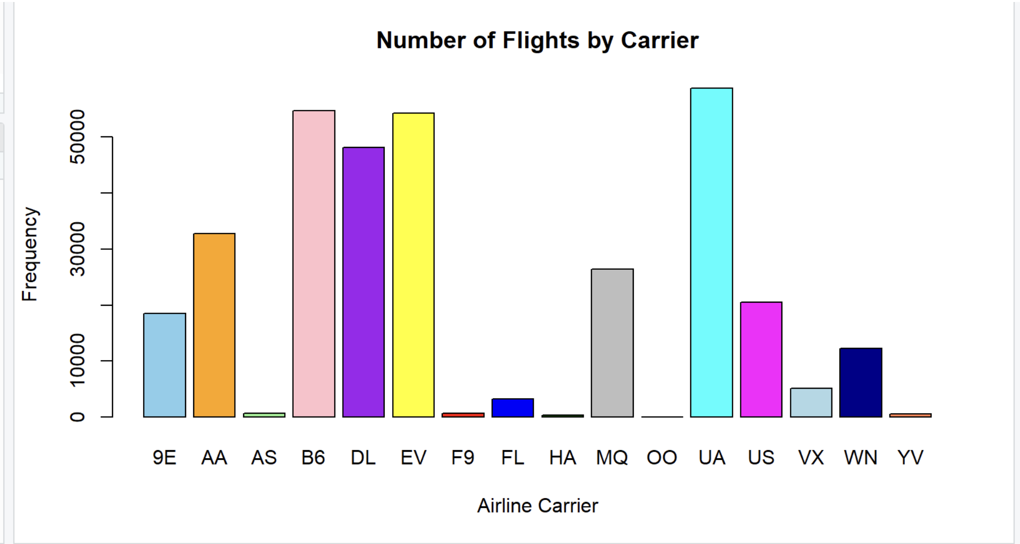

Figure 1: Number of Flights by Carrier

Inferences:

- United Airlines (UA) dominates with the highest number of departures from NYC
- JetBlue (B6) and ExpressJet (EV) are the second and third busiest carriers
- Several carriers such as HA, OO, and YV have very minimal presence at NYC airports

Boxplots

Departure Delay

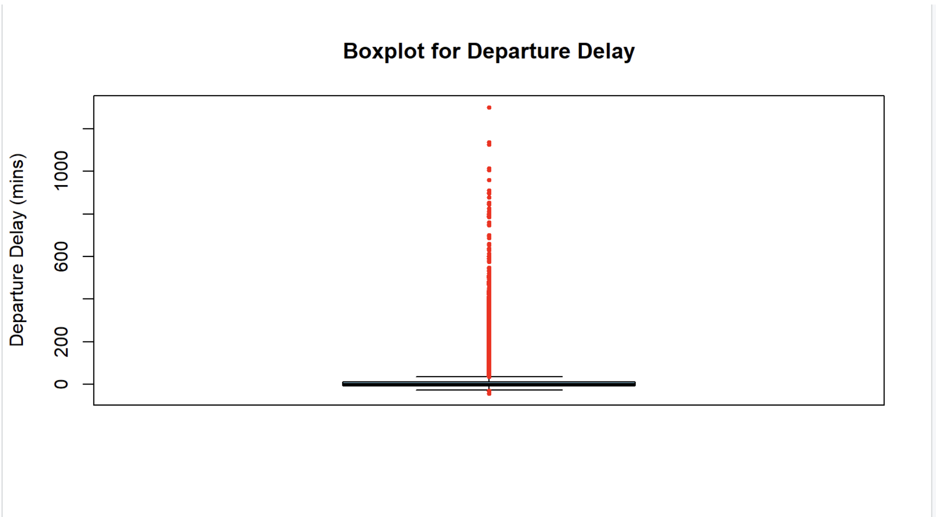

Figure 2: Boxplot for Departure Delay

Inferences:

- The box is compressed near zero with median at -2 minutes
- Massive number of red outlier points extending beyond 1,000 minutes
- Upper whisker far longer than lower whisker — extreme right skewness
- Most flights depart on time or slightly early

Arrival Delay

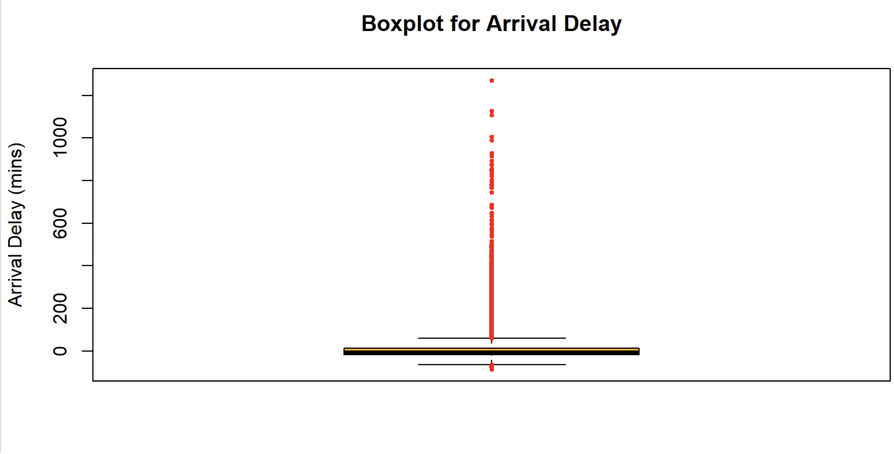

Figure 3: Boxplot for Arrival Delay

Inferences:

- Very similar pattern to departure delay with median at -5 minutes
- More than half of all flights arrive early
- Extreme outliers again reach beyond 1,000 minutes

Air Time

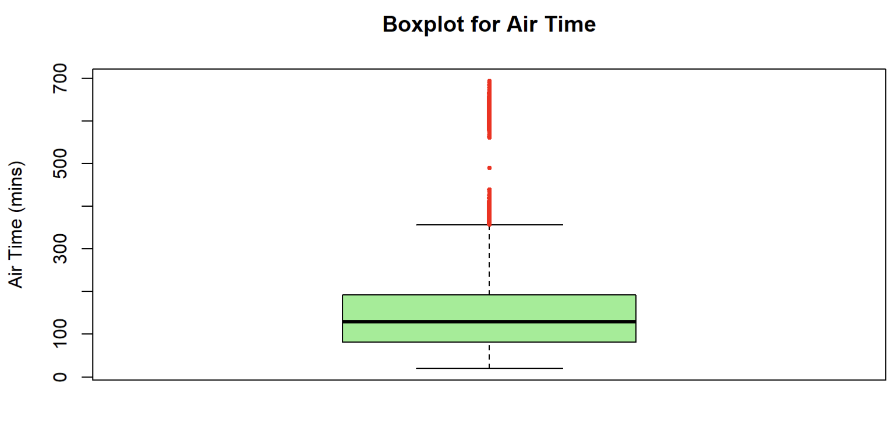

Figure 4: Boxplot for Air Time

Inferences:

- IQR spans approximately 80 to 200 minutes with median of 129 minutes
- Outliers above 400 minutes represent long-haul domestic flights
- More balanced and symmetric than the delay variables

Distance

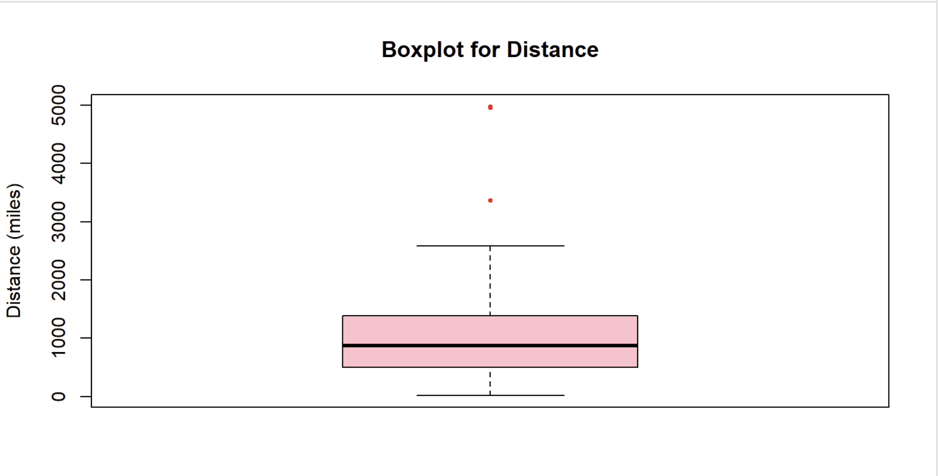

Figure 5: Boxplot for Distance

Inferences:

- IQR spans approximately 500 to 1,400 miles with median of 872 miles
- Only two extreme outliers visible — flights to Hawaii (≈5,000 miles)
- Distribution is moderately right skewed

Histograms

Departure Delay

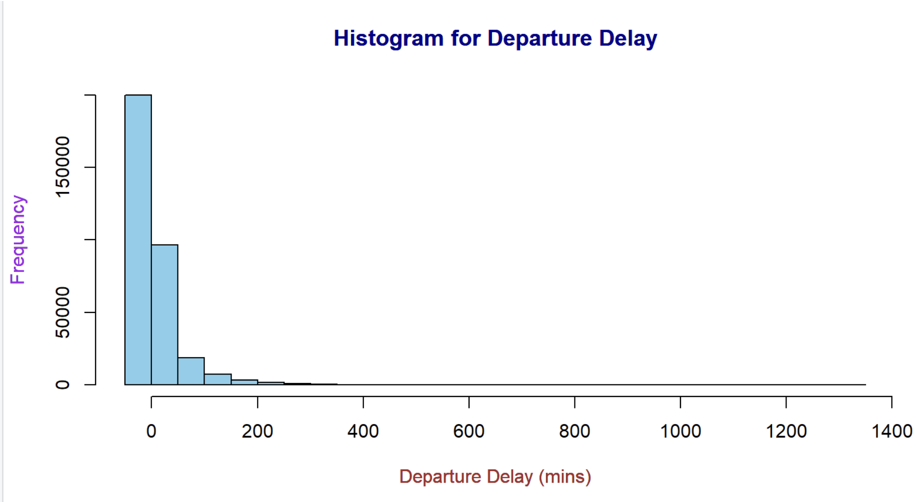

Figure 6: Histogram for Departure Delay

Inferences:

- Extremely tall bar near zero followed by a rapid drop
- Long thin right tail extending to 1,400 minutes
- Classic heavy right skew — majority of flights on time, few extremely late

Arrival Delay

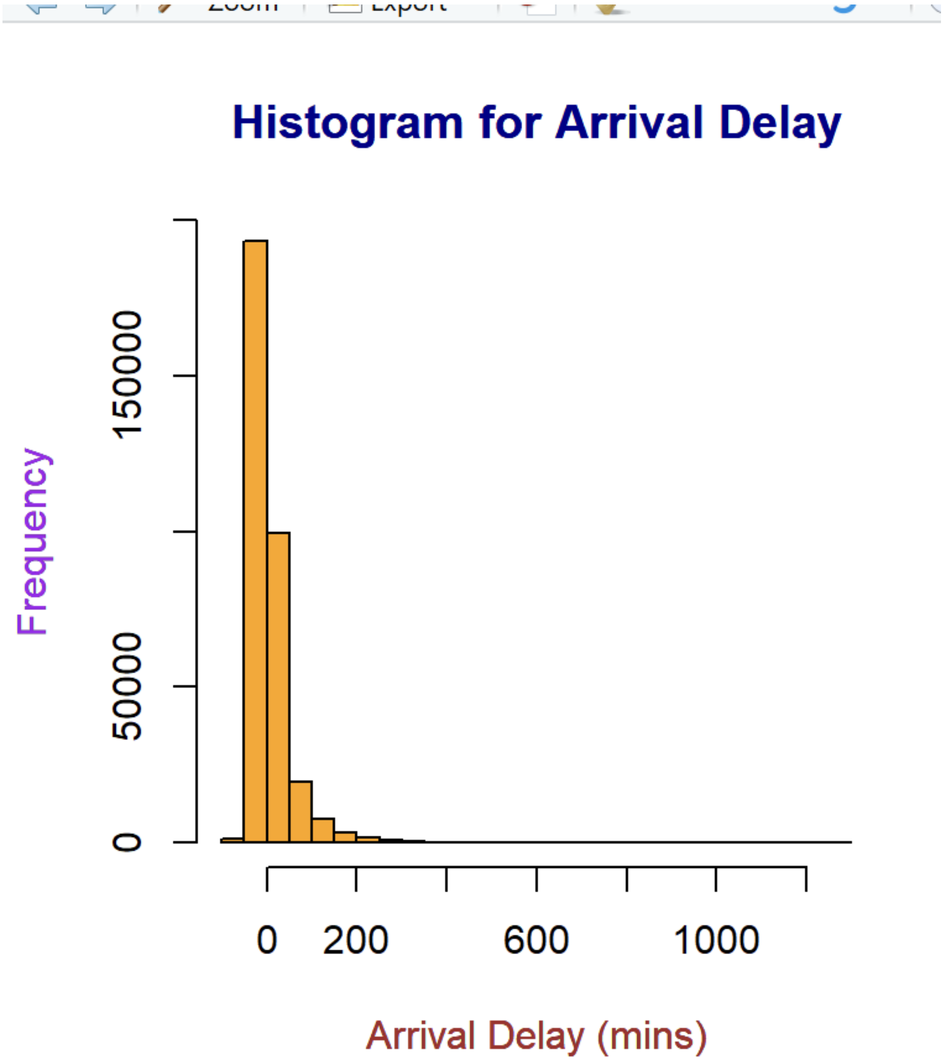

Figure 7: Histogram for Arrival Delay

Inferences:

- Nearly identical shape to departure delay histogram
- Peak near zero with long right tail

Air Time

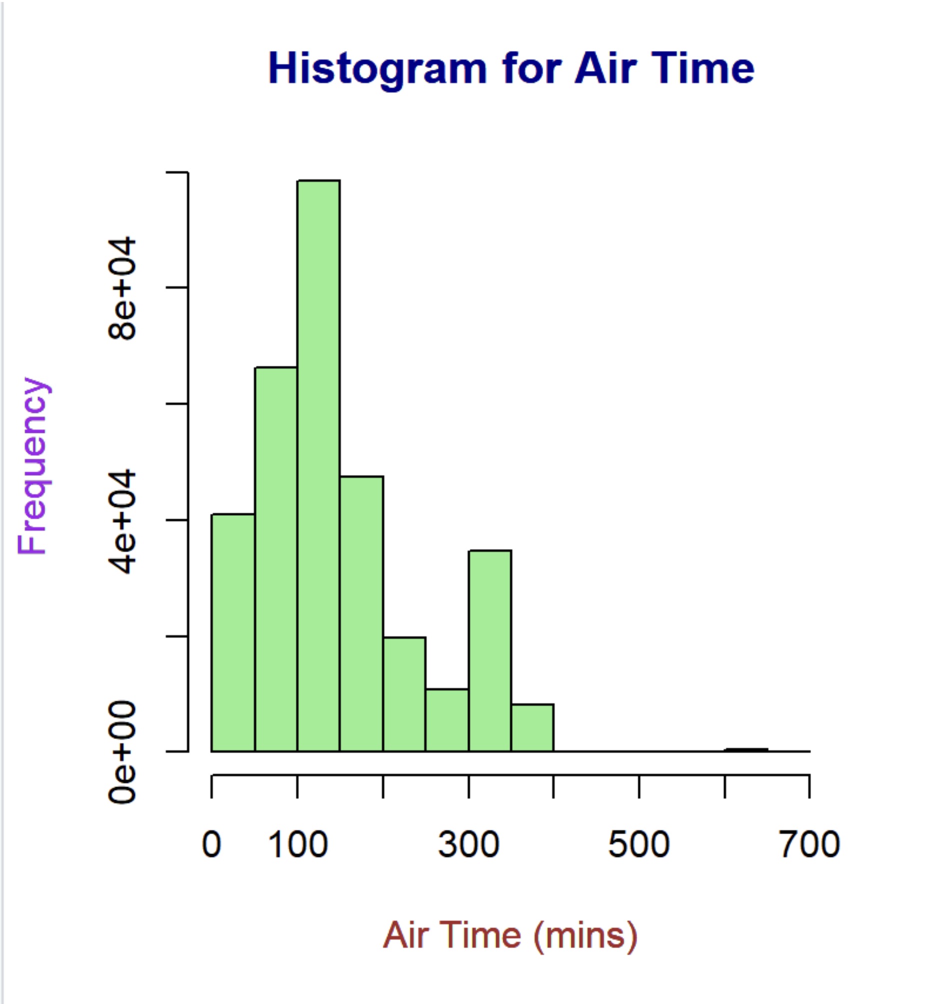

Figure 8: Histogram for Air Time

Inferences:

- Primary peak at 100–150 minutes (short flights)
- Secondary peak around 300 minutes (medium to long flights)

Distance

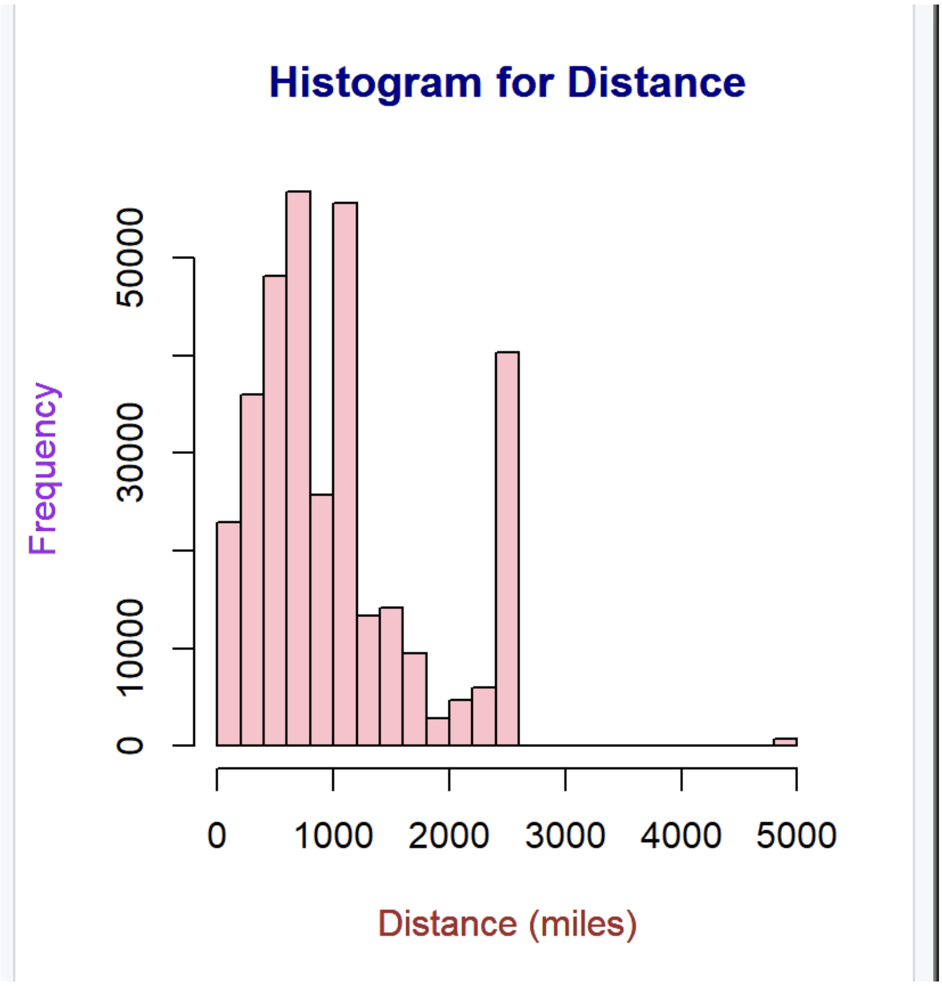

Figure 9: Histogram for Distance

Inferences:

- Dominant peak under 1,000 miles (short routes)
- Secondary peak around 2,500 miles
- Isolated bar at 5,000 miles corresponds to Hawaii flights

Having examined all four variables, departure delay was selected as the focus variable for further analysis as a separate dataset. It showed unique qualities such as extreme outliers and heavy right skewness. Analyzing it further through distribution fitting, MLE, and sufficiency would allow us to better understand and model the underlying behaviour of flight delays.

QQ Plots: Distribution Fitting for Departure Delay

Only positive departure delays were used since Exponential and Gamma distributions (that we are taking as sample assumption distributions for the dataset) are defined for positive values only. Four distributions were tested to identify the best fit.

Normal Distribution

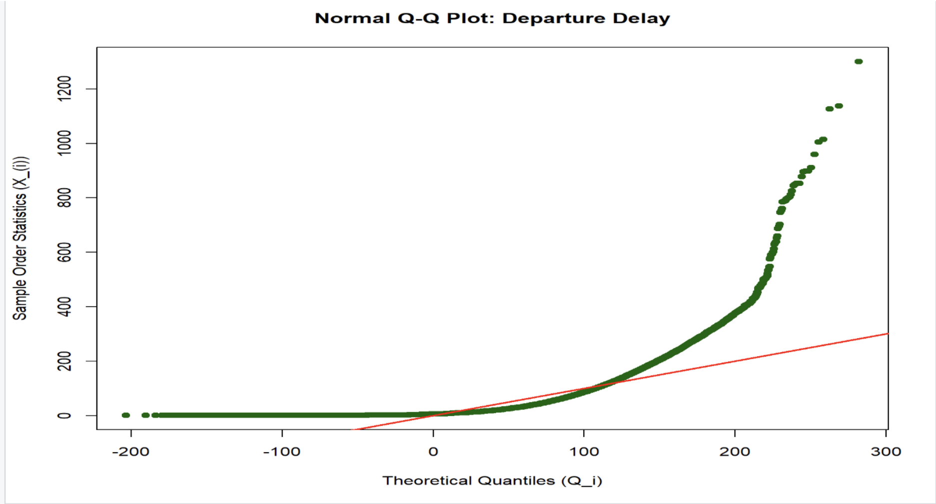

Figure 10: Normal Q-Q Plot: Departure Delay

Inferences:

- Points deviate severely from the reference line throughout
- Upward curve deviates further confirming heavy right tail
- **Normal distribution is not a suitable model** for departure delay

Exponential Distribution

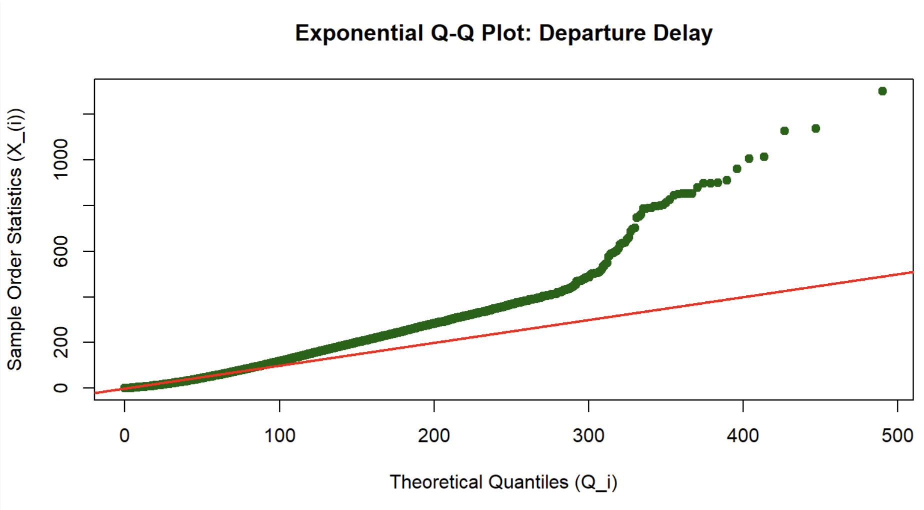

Figure 11: Exponential Q-Q Plot: Departure Delay

Inferences:

- Points follow the reference line in the lower part of the graph
- Sharp upward deviation after that

- **Partial fit only**

Chi-Square Distribution

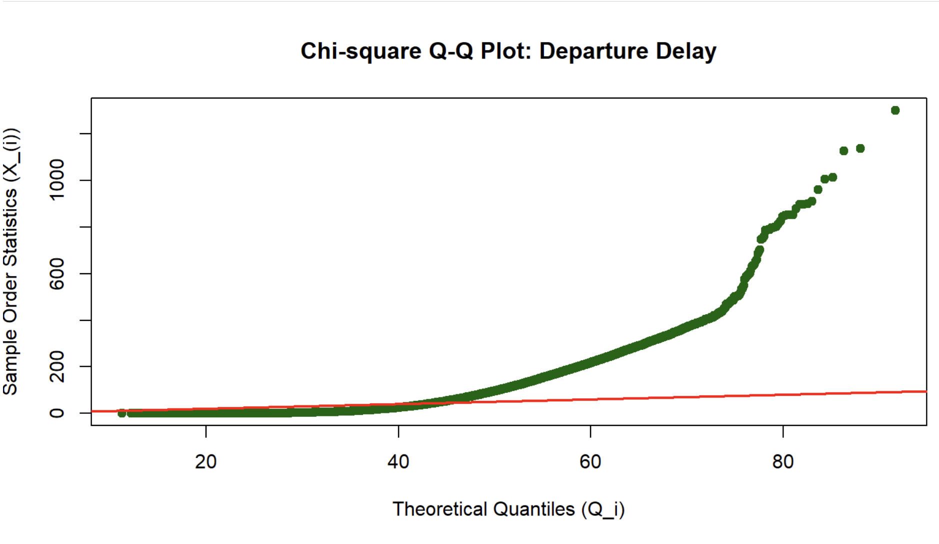

Figure 12: Chi-Square Q-Q Plot: Departure Delay

Inferences:

- Points remain flat near zero for a long stretch before rising sharply
- Very poor fit throughout
- **Chi-square distribution is not appropriate** for departure delay

Gamma Distribution

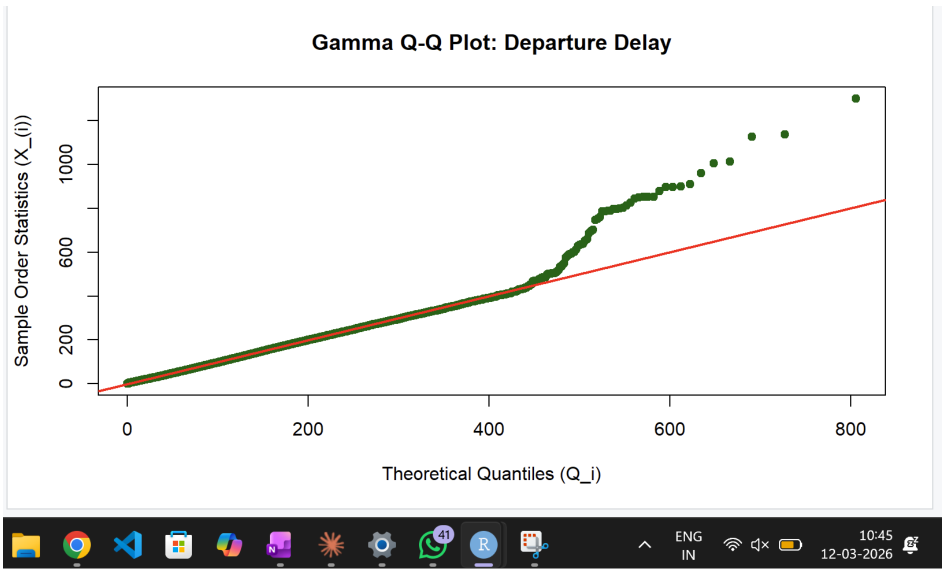

Figure 13: Gamma Q-Q Plot: Departure Delay

Inferences:

- Points follow the reference line closely from 0 to approximately 400 minutes
- Only the most extreme outlier delays deviate from the line
- **Gamma is the best fitting distribution** for positive departure delays

Table 3: Summary of QQ Plot Results

Distribution	Fit	Reason
Normal	Poor	Severe curve throughout
Exponential	Partial	Good lower range, poor upper tail
Chi-square	Poor	Flat then explosive deviation
Gamma	**Best**	Closely follows line across full range

Maximum Likelihood Estimation

Having confirmed Gamma as the best fitting distribution, MLE was used to estimate its parameters. A continuous random variable X follows a **Gamma distribution** with shape parameter $\alpha > 0$ and scale parameter $\beta > 0$, denoted $X \sim \text{Gamma}(\alpha, \beta)$, with PDF:

$$f_X(x) = \frac{1}{\beta^\alpha \, \Gamma(\alpha)} \, x^{\alpha-1} e^{-x/\beta}, \quad x > 0$$

where $E(X) = \alpha\beta$ and $\text{Var}(X) = \alpha\beta^2$.

The likelihood function for an i.i.d. sample $X_1, \ldots, X_n$ is defined as:

$$\mathcal{L}(\theta \mid \mathbf{x}) \stackrel{\text{def}}{=} \prod_{i=1}^{n} f_{X_i}(x_i;\, \theta), \quad \theta \in \Theta$$

The MLE $\hat{\theta}_{MLE}$ satisfies $\mathcal{L}(\hat{\theta}_{MLE} \mid \mathbf{x}) \geq \mathcal{L}(\theta \mid \mathbf{x})$ for all $\theta \in \Theta$, equivalently:

$$\hat{\theta}_{MLE} = \underset{\theta \in \Theta}{\arg\max} \sum_{i=1}^{n} \log\big(f_{X_i}(x_i;\, \theta)\big)$$

Note: R's `fitdistr()` uses the *rate* parameterization (rate $= 1/\beta$). The estimates below are in that form as returned by R.

Table 4: MLE Parameter Estimates — Gamma Distribution

Parameter	Estimate	Interpretation
Shape ($\hat{\alpha}$)	0.7210	< 1 confirms heavy right skew
Rate ($\widehat{1/\beta}$)	0.0183	Small value confirms wide spread
Mean ($\hat{\alpha}/\widehat{1/\beta}$)	39.40 mins	Average delay for delayed flights

Inferences:

- Shape < 1 is consistent with the heavy right skew seen in histograms and boxplots
- MLE mean of 39.40 minutes is the estimated average delay for actually delayed flights
- This is much higher than the overall mean of 12.64 minutes which includes early flights

Minimization of Negative Log-Likelihood

Maximising the likelihood is equivalent to minimising the negative log-likelihood:

$$\hat{\theta} = \arg\min_{\theta} \big[-\log \mathcal{L}(\theta)\big] = -\sum_{i=1}^{n} [\alpha \log \beta - \log \Gamma(\alpha) + (\alpha - 1) \log x_i - \beta x_i]$$

The negative log-likelihood was minimised using `optim()` in R and results compared with `fitdistr()`:

Table 5: MLE vs Minimization Results

Parameter	MLE	Minimization	Difference
Shape (α)	0.720997	0.720868	0.000129
Rate ($1/\beta$)	0.018300	0.018309	0.000009

Inferences:

- Both methods produce virtually identical results
- Tiny differences are due to numerical rounding only
- Confirms that maximising likelihood = minimising negative log-likelihood
- Validates the MLE parameter estimates obtained earlier

Minimal Sufficient Statistics

Concept

A statistic $T = T(X_1, X_2, \ldots, X_n)$ is said to be **sufficient** for $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ if the conditional distribution of the sample $\mathbf{X} = (X_1, \ldots, X_n)$ given $T = t$ does not depend on θ.

A statistic $T(\mathbf{X})$ is **minimal sufficient** for θ if:

- $T(\mathbf{X})$ is sufficient for θ, and
- For any other sufficient statistic $S(\mathbf{X})$, T is a function of S (i.e., $T = g(S)$ for some function g).

The minimal sufficient statistic achieves the greatest possible data reduction while retaining all information about θ.

Minimal Sufficiency for Gamma Distribution

The Gamma PDF can be written in exponential family form as:

$$f(x;\,\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\,x^{\alpha-1}e^{-\beta x} = \frac{\beta^\alpha}{\Gamma(\alpha)}\,\exp\bigl((\alpha-1)\ln x - \beta x\bigr)$$

By the **Minimal Sufficiency Theorem for the Exponential Family**, the minimal sufficient statistic is:

$$\mathbf{T}(\mathbf{X}) = \left(\sum_{i=1}^{n} \ln X_i,\ \ \sum_{i=1}^{n} X_i\right)$$

Since $\sum_{i=1}^{n} X_i = n\bar{X}$ is a one-to-one function of $\bar{X}$, this is equivalently written as:

$$\mathbf{T}(\mathbf{X}) = \left(\sum_{i=1}^{n} \ln X_i,\ \ \bar{X}\right)$$

Table 6: Minimal Sufficient Statistics for Positive Departure Delays

Statistic	**Value**	**Role**
$\sum_{i=1}^{n} \ln X_i$	−39,847.2	Sufficient for shape α
Sample Mean $\bar{X}$	39.373 mins	Sufficient for rate $1/\beta$

Inferences:

- The entire dataset of 100,000+ delay observations reduces to just **two numbers**
- These two values capture **all** statistical information about α and β
- No other statistic can provide more information about the Gamma parameters
- The minimal sufficient statistic follows directly from the exponential family form of the Gamma distribution
- MLE estimators are functions of these minimal sufficient statistics — confirming MLE is optimal

Conclusion

- The nycflights13 dataset contains 336,776 flight records with 19 variables across 3 NYC airports
- Variables were classified into 4 qualitative and 7 quantitative types
- Departure and arrival delays are **heavily right skewed** — most flights are on time but a few are severely delayed

- United Airlines (UA) dominates NYC departures; EWR is the busiest origin airport
- Histograms and boxplots confirmed extreme right skewness in delays
- QQ plot testing confirmed that **positive departure delays follow a Gamma distribution**
- MLE estimated Gamma parameters: Shape = 0.721, Rate = 0.018, Mean delay = 39.40 mins
- Minimisation of the negative log-likelihood independently confirmed the MLE estimates
- The minimal sufficient statistic $\left(\sum \ln X_i,\ \bar{X}\right)$ compresses all delay information into two numbers without any loss of information about the Gamma parameters

Using the fitted Gamma model, we can predict the probability of a delay exceeding any threshold. For example, where F is the Gamma CDF with $\hat{\alpha} = 0.721$ and rate = 0.018:

$$P(X > 60) = 1 - F(60) = 0.2209$$
$$P(X > 120) = 1 - F(120) = 0.0644$$
$$P(20 < X < 60) = F(60) - F(20) = 0.3211$$

This means roughly 1 in 5 delayed flights exceed one hour, and about 1 in 3 delayed flights fall in the 20–60 minute range. Such predictions can help airlines set buffer times and help passengers assess the risk of missing connecting flights.